

SCHEDULE OF QUANTITIES FOR PART-A Electrical WORKS - CE-1001 LOT-4(ADDENDUM)

S.No.	Description	Units	IGD Airport	mid shaft	Palam	Dashrath Puri	Dabri Mor	Janakpuri West	Total Quantity	Rate in INR	Rate in Foreign Currency	Amount in INR	Amount in Foreign Currency
g)	Emergency stop push button lockable type.												
h)	LED indication lamp for ON, OFF, Trip.												
i)	3 Nos with suitable rating AC-3 duty, 3 Pole Contactor with Auxiliary												
j)	One no star-delta timer.												
k)	1 No. Auxiliary Contactor with 4 Nos. Auxiliary Contacts for Remote												
l)	1 No Single Phasing Preventor Relay.												
m)	Power & control terminals for cable connections.												
n)	Provision for terminals for wiring to Local control panel (LCP)												
o)	Digital Energy meter.												
5)	8 Sets of Star - Delta Starter unit of 11 kW and each shall be equipped with the followings but not limited to:(For CT-001-F1, CT-001-F2,CT-002-F1,CT-002-F2,CT-003-F1,CT-003-F2 & 2 no spare)												
a)	1 No. 3 Pole 25 Amps MPCB (with Under Voltage Protection and inbuilt Single Phasing Preventor) with 20-25 A Over - load release and having contacts for indications - On, Off, Trip.												
b)	1 No. Digital Ammeter, CT operated.												
c)	1 No. CT 25/5A, 15VA, class - I for metering.												
d)	Local / Remote Selector Switch key operated.												
e)	Door mounted push buttons for Start and Stop functions.												
f)	Emergency stop push button lockable type.												
g)	LED indication lamp for ON, OFF, Trip.												
h)	3 No. with suitable rating AC-3 Duty, 3 - pole Contactor with Auxiliary Contacts and Operating Coil of 230 V / 110 V AC.(for Main, Star & Delta Contactor).												
i)	One no Star-Delta timer.												
j)	1 No. Auxiliary Contactor with 4 Nos. Auxiliary Contacts for Remote Operation.												
k)	Power & control terminals for cable connections.												
l)	Provision for terminals for wiring to Local control panel (LCP)												
m)	Digital Energy meter.												
6)	2 nos outgoing feeders for Capacitor Panel with each consisting of the following												
a)	1 No. 800 A, 415V, 65 50kA, 3 Pole, Electrically operated Draw Out type, ACB complete with safety shutters, 240V AC or 110V DC (control supply voltage to be selected in consultation with BMS Contractor) shunt trip coil & closing coil and having microprocessor based releases with (1) Protection against long time + short time + instantaneous + earth fault (all with adjustable time delays) (2) NO / NC contacts for interlocks and indications (3) 1 no CT in neutral to connect with the trip unit.												
b)	TNC switch												
c)	Selector switch for auto/manual operation.												
d)	LED indication lamp for ON, OFF & TRIP.												
e)	Digital Ammeter with Inbuilt Selector Switch												
f)	3 Nos. CTs 800/5A, 15VA, CL-1 for metering.												
g)	Power & control terminals for cable connections.												
7)	4No. 100 A, 415V, 50KA, 3P MCCB having Electronic Trip Unit with adjustable protection against Overload, adjustable protection against Short Circuit.(For DB-330, LDB-AB, SPDB-AB & 1NO. SPARE).												
a)	1 No. Digital Ammeter, CT operated.												
b)	1 No. CT 100/5A, 15VA, class - I for metering.												
c)	LED indication lamp for ON, OFF, Trip.												

SCHEDULE OF QUANTITIES FOR PART-A Electrical WORKS - CE-16211 LOT-4(ADDENDUM)

S.No.	Description	Units	IGD Airport	mid shaft	Palani	Dashrath Puri	Dabri Mor	Janakpuri West	Total Quantity	Rate in INR	Rate in Foreign Currency	Amount in INR	Amount in Foreign Currency
8)	4 No. 400 A, 415V, 50KA, 3P MCCB having Electronic Trip Unit for Fire Pump Panel, Air Cooled Chiller Panel, Pump Room Panel & 1 no. spare with adjustable protection against Overload, adjustable protection against Short Circuit.												
a)	1 No. Digital Ammeter, CT operated.												
b)	1 No. CT 400/5A, 15VA, class - I for metering.												
c)	LED indication lamp for ON, OFF, Trip.												
1.01.2	Delivery of equipment to site	set			1	1	1		3				
1.01.3	Installation and Equipment testing	set			1	1	1		3	50000		150000	
1.01.4	System testing, Commissioning and hand over	set			1	1	1		3	3780		11340	
1.01.5	Interlocking Arrangement	set			1	1	1		3	3150		9450	
1.01.6	Scada BMS Connectivity at the separate Marshalling box in the panel with provision of all necessary I/O ports & transducers mounting prewired upto Marshalling box terminal boards from the factory itself and audited in factory from the BMS provider.	set			1	1	1		3	1890		5670	
1.02	MAIN PLANT PANEL DB-170/270 (Type-2 for chiller plant room with 4 chillers)												
	Panel complete as per specifications, drawings, requirements & following details:	Sets	1					1	2	9478780		18957560	
A	TRANSFORMER - III INCOMING shall be equipped with												
a)	4 2No. 3200 4000A, 415V, 65 56KA, 4 Pole, Electrically operated Draw Out type, ACB complete with safety shutters, 240V AC shunt trip coil & closing coil and having microprocessor based releases with (1) Communication capability (2) Provision for measurement of three phase current (3) Protection against long time + short time + earth fault (all with adjustable time delays) + instantaneous (4) Zone Selective Interlocking Unit for Total Discrimination (5) Communication options to remotely read and set parameters for the protection functions, transmission of ammeter measurements (6) Signalling of the cause of tripping and alarms, maximeter reset etc (7) NO / NC contacts for interlocks and indications (8) 1 no CT in neutral to connect with the trip unit.												
b)	TNC switch												
c)	Auto / Local / Remote Selector Switch key operated.												
d)	LED indication lamp for ON, OFF & TRIP & Phase indication R-Y-B.												
e)	Metering												
i)	Multi function energy meter for measurement of voltage, line current, power factor, power & energy.												
ii)	4 Nos. CTs 3200/5A, 15VA, CL-1 for metering.												
iii)	1 Nos. CTs 3200/5A, 15VA, CL-1 for APFCR relay.												
B	ASS 3 TO ASS 1 (THE BREAKER for connecting DB-170/270 to DB-100)												

SCHEDULE OF QUANTITIES FOR PART-A Electrical WORKS - CE-10 LOT-4(ADDENDUM)

S.No.	Description	Units	IGD Airport	mid shaft	Palam	Dashrath Puri	Debri Mor	Janakpuri West	Total Quantity	Rate in INR	Rate in Foreign Currency	Amount in INR	Amount in Foreign Currency
a)	1 No. 3200A, 415V, 65kA, 4 Pole, Electrically operated Draw Out type, ACB complete with safety shutters, 240V AC shunt trip coil & closing coil and having microprocessor based releases with (1) Communication capability (2) Provision for measurement of three phase and neutral current (3) Protection against long time + short time + earth fault (all with adjustable time delays) + instantaneous (4) Zone Selective Interlocking Unit for Total Discrimination (5) Communication options to remotely read and set parameters for the protection functions, transmission of ammeter measurements (6) Signalling of the cause of tripping and alarms, maximeter reset etc (7) NO / NC contacts for interlocks and indications. (8) Status indication-ON, OFF, TRIP.												
b)	TNC-switch												
c)	Auto / Local / Remote Selector Switch, Key operated.												
d)	LED indication lamp for ON, OFF & TRIP.												
e)	Digital Ammeter with Inbuilt Selector Switch												
f)	3 Nos. CTs, 3200/5A, 15VA, CL-1 for metering.												
C)	BUS COUPLER												
a)	1 Nos. 2500A, 415 V, 56KA, 50 Hz TPN EDO type ACBs interlocked with buscoupler and complete with microprocessor based releases having SCADA compatability.												
b)	Terminals to receive XLPE armoured power & control cables												
c)	Digital Ammeter with Inbuilt Selector Switch												
d)	3 Nos. CTs, 2500/5A, 15VA, CL-1 for metering.												
D	BUSBAR												
	Electrolytic high conductivity tinned copper three phase and neutral busbars rated at 3200 A having a maximum current density of 1.2 A per sqmm suitable to withstand symmetrical fault level of 56 kA at 415 V.												
E	OUT GOING FEEDERS												
1	4 Sets feeder for Chiller Motors up to 276 KW each shall be equipped with the followings but not limited to:												
a)	1 No. 800 A, 415V, 65 56kA, 3 Pole, Electrically operated Draw Out type, ACB complete with safety shutters, 240V AC or 110V DC (control supply voltage to be selected in consultation with BMS Contractor) shunt trip coil & closing coil and having microprocessor based releases with (1) Protection against long time + short time + instantaneous + earth fault (all with adjustable time delays) (2) NO / NC contacts for interlocks and indications (3) 1 no CT in neutral to connect with the trip unit.												
b)	TNC switch												
c)	Selector switch for auto/manual operation.												
d)	LED indication lamp for ON, OFF & TRIP.												
e)	Digital Ammeter with Inbuilt Selector Switch												
f)	3 Nos. CTs 800/5A, 15VA, CL-1 for metering.												
g)	Digital Energy meter.												
h)	Power & control terminals for cable connections.												
2	5 Sets of Star - Delta Starter Unit for Primary Chilled Water Pump Motors of 12 kW (4 Working + 1 Standby) and each shall be equipped with the followings but not limited to: (For -CHWP-001P, CHWP-002P, CHWP-003P, CHWP-04P, CHWP-05P)												

SCHEDULE OF QUANTITIES FOR PART-A Electrical WORKS - CE-10 and 11 LOT-4(ADDENDUM)

S.No.	Description	Units	IGD Airport	mid shaft	Palam	Dashrath Puri	Dabri Mor	Janakpuri West	Total Quantity	Rate in INR	Rate in Foreign Currency	Amount in INR	Amount in Foreign Currency
a)	1 No. 3 Pole 100A, 50 56kA at 415V, 3P motor duty MCCB with Electronic Trip Unit(with inbuilt over load and short circuit protection) over load range according to motor selection, and auxiliary Contacts.												
b)	1 No. Digital Ammeter, CT operated.												
c)	1 No. CT 100/5A, 15VA, class - I for metering.												
d)	Local / Remote Selector Switch key operated.												
e)	Working / Standby Selector Switch key operated.												
f)	Door mounted push buttons for Start and Stop functions.												
g)	Emergency stop push button lockable type.												
h)	LED indication lamp for ON, OFF, Trip.												
i)	3 Nos suitable rating AC-3 duty, 3 Pole Contactor with Auxiliary Contacts, Operating coil 230V / 110V AC (for Main, Star & Delta Contactor).												
j)	One no star-delta timer.												
k)	1 No. Auxiliary Contactor with 4 Nos. Auxiliary Contacts for Remote Operation.												
l)	1 No Single Phasing Preventor Relay.												
m)	Power & control terminals for cable connections.												
n)	Digital Energy meter.												
o)	Provision for terminals for wiring to Local control panel (LCP)												
3	4 Nos. Feeder for Secondary Chilled Water Pump Motors of 37 kW (3 Working + 1 Standby) and each shall be equipped with the followings but not limited to:(For CHWP-001S, CHWP-002S, CHWP-003S, CHWP-004S)												
a)	1 No. 3 Pole 100A, 50 56kA at 415V, 3P motor duty MCCB with Electronic Trip Unit(with inbuilt over load and short circuit protection) over load range according to motor selection, and auxiliary Contacts.												
b)	1 No. Digital Ammeter, CT operated.												
c)	1 No. CT 100/5A, 15VA, class - I for metering.												
d)	LED indication lamp for ON, OFF, Trip.												
e)	Digital Energy meter.												
4)	56 Sets of Star - Delta Stater unit for Condenser Water Pump Motors up to 37 kW (4 Working + 1 Standby) and each shall be equipped with the followings but not limited to:(For CDWP-001, CDWP-002, CDWP-003, CDWP-04, CDWP-05)												
a)	1 No. 3 Pole 100A, 50 56kA at 415V, 3P motor duty MCCB with Electronic Trip Unit(with inbuilt over load and short circuit protection) over load range to suit actual pump selection, and auxiliary Contacts.												
b)	1 No. Digital Ammeter, CT operated.												
c)	1 No. CT 100/5A, 15VA, class - I for metering.												
d)	Local / Remote Selector Switch key operated.												
e)	Working / Standby Selector Switch key operated.												
f)	Door mounted push buttons for Start and Stop functions.												
g)	Emergency stop push button lockable type.												
h)	LED indication lamp for ON, OFF, Trip.												
i)	3 Nos with suitable rating AC-3 duty, 3 Pole Contactor with Auxiliary Contacts, Operating coil 230V / 110V AC (for Main, Star & Delta Contactor).												
j)	One no star-delta timer.												
k)	1 No. Auxiliary Contactor with 4 Nos. Auxiliary Contacts for Remote Operation.												
l)	1 No Single Phasing Preventor Relay.												
m)	Power & control terminals for cable connections.												
n)	Digital Energy meter.												
o)	Provision for terminals for wiring to Local control panel (LCP)												

SCHEDULE OF QUANTITIES FOR PART-A Electrical WORKS - CE-10 and 11 LOT-4(ADDENDUM)

S.No.	Description	Units	IGD Airport	mid shaft	Palam	Dashrath Puri	Dabri Mor	Janakpuri West	Total Quantity	Rate in INR	Rate in Foreign Currency	Amount in INR	Amount in Foreign Currency
5)	10 Sets of Star - Delta Starter unit of 11 kW and each shall be equipped with the followings but not limited to: (For CT-001-F1, CT-001-F2, CT-002-F1, CT-002-F2, CT-003-F1, CT-003-F2, CT-004-F1, CT-004-F2 & 2 no spare)												
a)	1 No. 3 Pole 25 Amps MPCB (with Under Voltage Protection and inbuilt Single Phasing Preventor) with 20-25 A Over - load release and having contacts for indications - On, Off, Trip.												
b)	1 No. Digital Ammeter, CT operated.												
c)	1 No. CT 25/5A, 15VA, class - I for metering.												
d)	Local / Remote Selector Switch key operated.												
e)	Door mounted push buttons for Start and Stop functions.												
f)	Emergency stop push button lockable type.												
g)	LED indication lamp for ON, OFF, Trip.												
h)	3 No. with suitable rating AC-3 Duty, 3 - pole Contactor with Auxiliary Contacts and Operating Coil of 230 V / 110 V AC. (for Main, Star & Delta Contactor).												
i)	One no Star-Delta timer.												
j)	1 No. Auxiliary Contactor with 4 Nos. Auxiliary Contacts for Remote Operation.												
k)	Power & control terminals for cable connections.												
l)	Digital Energy meter.												
m)	Provision for terminals for wiring to Local control panel (LCP)												
6)	1 nos outgoing feeders for Capacitor Panel with each consisting of the following												
a)	1 No. 1250 A, 415V, 65 56 kA, 3 Pole, Electrically operated Draw Out type, ACB complete with safety shutters, 240V AC or 110V DC (control supply voltage to be selected in consultation with BMS Contractor) shunt trip coil & closing coil and having microprocessor based releases with (1) Protection against long time + short time + instantaneous + earth fault (all with adjustable time delays) (2) NO / NC contacts for interlocks and indications (3) 1 no CT in neutral to connect with the trip unit.												
b)	TNC switch												
c)	Selector switch for auto/manual operation.												
d)	LED indication lamp for ON, OFF & TRIP.												
e)	Digital Ammeter with Inbuilt Selector Switch												
f)	3 Nos. CTs 1250/5A, 15VA, CL-1 for metering.												
g)	Power & control terminals for cable connections.												
7)	8Nos. 100 A, 415V, 50 56 KA, 3P MCCB having Electronic Trip Unit with adjustable protection against Overload, adjustable protection against Short Circuit. (For DB-330, LDBs, SPDBs & 2Nos. Spare).												
a)	1 No. Digital Ammeter, CT operated.												
b)	1 No. CT 100/5A, 15VA, class - I for metering.												
c)	LED indication lamp for ON, OFF, Trip.												
8)	4 Nos. 400 A, 415V, 50 56 KA, 3P MCCB having Electronic Trip Unit for Fire Pump Panel, Air Cooled Chiller Panel, Pump Room Panel & 1 No. spare with adjustable protection against Overload, adjustable protection against Short Circuit.												

SCHEDULE OF QUANTITIES FOR PART-A Electrical WORKS - CE-10 and 11 LOT-4(ADDENDUM)

S.No.	Description	Units	IGD Airport	mid shaft	Palam	Dashrath Pur	Dabri Mor	Janakpur West	Total Quantity	Rate in INR	Rate in Foreign Currency	Amount in INR	Amount in Foreign Currency
a)	1 No. Digital Ammeter, CT operated.												
b)	1 No. CT 400/5A, 15VA, class - I for metering.												
c)	LED indication lamp for ON, OFF, Trip.												
1.02.2	Delivery of equipment to site	set	1					1	2				
1.02.3	Installation and Equipment testing	set	1					1	2	50000		100000	
1.02.4	System testing, Commissioning and hand over	set	1					1	2	3780		7560	
1.02.5	Interlocking Arrangement	set	1					1	2	3150		6300	
1.02.6	Scada\ BMS Connectivity at the separate Marshalling box in the panel with provision of all necessary I/O points & transducers mounting prewired upto Marshalling box terminal boards from the factory itself and audited in factory from the BMS provider.	set	1					1	2	1890		3780	
1.03	MAIN PLANT PANEL DB-170/270 (Type-3 for chiller plant room with 4 chillers)	Sets								8854680			
	Panel complete as per specifications, drawings, requirements & following details:												
A	INCOMERS												
1)	2 Nos. 2500A, 415 V, 65 50KA, 50 Hz TPN EDO type ACBs interlocked with buscoupler and complete with microprocessor based releases having SCADA compatability and energy metering.												
2)	2 sets of Red (on), Green (off) and amber (C.B. Fault) indicating lamps												
3)	2 sets of (three) white phase indicating lamps												
4)	Metering												
i)	2 Nos. Multifunction energy meter for measurement of voltage, line current, power factor, power & energy.												
ii)	3 Nos. CTs 2500/5A, 15VA, CL-1 for metering.												
5)	Terminals to receive XLPE armoured power & control cables												
B	BUSBAR												
	2 Nos. Electrolytic high conductivity tinned copper three phase and neutral busbars rated at 2500 A having a maximum current density of 1.2 A per sqmm suitable to withstand symmetrical fault level of 56 50kA at 415 V.												
C	BUS COUPLER												
1)	1 Nos. 2500A, 415 V, 65 50KA, 50 Hz TPN EDO type ACBs interlocked with buscoupler and complete with microprocessor based releases having SCADA compatability.												
2)	Terminals to receive XLPE armoured power & control cables												
3)	Digital Ammeter with Inbuilt Selector Switch												
4)	3 Nos. CTs, 2500/5A, 15VA, CL-1 for metering.												
D	OUT GOING FEEDERS												
1	4 Sets feeder for Chiller Motors up to 276 KW each shall be equipped with the followings but not limited to:												
a)	1 No. 800 A, 415V, 65 50kA, 3 Pole, Electrically operated Draw Out type, ACB complete with safety shutters, 240V AC or 110V DC (control supply voltage to be selected in consultation with BMS Contractor) shunt trip coil & closing coil and having microprocessor based releases with (1) Protection against long time + short time + instantaneous + earth fault (all with adjustable time delays) (2) NO / NC contacts for interlocks and indications (3) 1 no CT in neutral to connect with the trip unit.												

SCHEDULE OF QUANTITIES FOR PART-A Electrical WORKS - CE-16.1 LOT-4(ADDENDUM)

S.No.	Description	Units	IGD Airport	mid shaft	Palam	Dashrath Puri	Dabri Mor	Janakpuri West	Total Quantity	Rate in INR	Rate in Foreign Currency	Amount in INR	Amount in Foreign Currency
b)	TNC switch												
c)	Selector switch for auto/manual operation.												
d)	LED indication lamp for ON, OFF & TRIP.												
e)	Digital Ammeter with Inbuild Selector Switch												
f)	3 Nos. CTs 800/5A, 15VA, CL-1 for metering.												
g)	Digital Energy meter.												
h)	Power & control terminals for cable connections.												
2	5 Sets of Star - Delta Starter Unit for Primary Chilled Water Pump Motors of 12 kW (4 Working + 1 Standby) and each shall be equipped with the followings but not limited to:(For -CHWP-001P, CHWP-002P, CHWP-003P, CHWP-04P,CHWP-05P)												
a)	1 No. 3 Pole 100A, 50 kA at 415V, 3P motor duty MCCB with Electronic												
b)	1 No. Digital Ammeter, CT operated.												
c)	1 No. CT 100/5A, 15VA, class - I for metering.												
d)	Local / Remote Selector Switch key operated.												
e)	Working / Standby Selector Switch key operated.												
f)	Door mounted push buttons for Start and Stop functions.												
g)	Emergency stop push button lockable type.												
h)	LED indication lamp for ON, OFF, Trip.												
i)	3 Nos suitable rating AC-3 duty, 3 Pole Contactor with Auxiliary Contacts, Operating coil 230V / 110V AC (for Main, Star & Delta Contactor).												
j)	One no star-delta timer.												
k)	1 No. Auxiliary Contactor with 4 Nos. Auxiliary Contacts for Remote Operation.												
l)	1 No Single Phasing Preventor Relay.												
m)	Power & control terminals for cable connections.												
n)	Digital Energy meter.												
o)	Provision for terminals for wiring to Local control panel (LCP)												
3	4 Nos. Feeder for Secondary Chilled Water Pump Motors of 37 kW (3 Working + 1 Standby) and each shall be equipped with the followings but not limited to:(For CHWP-001S, CHWP-002S, CHWP-003S, CHWP-004S)												
a)	1 No. 3 Pole 100A, 50kA at 415V, 3P motor duty MCCB with Electronic												
b)	1 No. Digital Ammeter, CT operated.												
c)	1 No. CT 100/5A, 15VA, class - I for metering.												
d)	LED indication lamp for ON, OFF, Trip.												
e)	Digital Energy meter.												
4)	5 6 Sets of Star - Delta Stater unit for Condenser Water Pump Motors up to 37 kW (4 Working + 1 Standby) and each shall be equipped with the followings but not limited to:(For CDWP-001, CDWP-002, CDWP-003, CDWP-04 , CDWP-05)												
a)	1 No. 3 Pole 100A, 50kA at 415V, 3P motor duty MCCB with Electronic												
b)	1 No. Digital Ammeter, CT operated.												
c)	1 No. CT 100/5A, 15VA, class - I for metering.												
d)	Local / Remote Selector Switch key operated.												
e)	Working / Standby Selector Switch key operated.												
f)	Door mounted push buttons for Start and Stop functions.												
g)	Emergency stop push button lockable type.												
h)	LED indication lamp for ON, OFF, Trip.												
i)	3 Nos with suitable rating AC-3 duty, 3 Pole Contactor with Auxiliary												
j)	One no star-delta timer.												
k)	1 No. Auxiliary Contactor with 4 Nos. Auxiliary Contacts for Remote												

SCHEDULE OF QUANTITIES FOR PART-A Electrical WORKS - CE-10...1 LOT-4(ADDENDUM)

S.No.	Description	Units	IGD Airport	Ind shaft	Palam	Dashrath Puri	Dabri Mor	Janakpuri West	Total Quantity	Rate in INR	Rate in Foreign Currency	Amount in INR	Amount in Foreign Currency
l)	1 No Single Phasing Preventor Relay.												
m)	Power & control terminals for cable connections.												
n)	Provision for terminals for wiring to Local control panel (LCP)												
o)	Digital Energy meter.												
5)	10 Sets of Star - Delta Starter unit of 11 kW and each shall be equipped with the followings but not limited to:(For CT-001-F1, CT-001-F2,CT-002-F1,CT-002-F2,CT-003-F1,CT-003-F2, CT-004-F1,CT-004-F2 & 2 no spare)												
a)	1 No. 3 Pole 25 Amps MPCB (with Under Voltage Protection and inbuilt												
b)	1 No. Digital Ammeter, CT operated.												
c)	1 No. CT 25/5A, 15VA, class - I for metering.												
d)	Local / Remote Selector Switch key operated.												
e)	Door mounted push buttons for Start and Stop functions.												
f)	Emergency stop push button lockable type.												
g)	LED indication lamp for ON, OFF, Trip.												
h)	3 No. with suitable rating AC-3 Duty, 3 - pole Contactor with Auxiliary												
i)	One no Star-Delta timer.												
j)	1 No. Auxiliary Contactor with 4 Nos. Auxiliary Contacts for Remote												
k)	Power & control terminals for cable connections.												
l)	Digital Energy meter.												
m)	Provision for terminals for wiring to Local control panel (LCP)												
6)	1 nos outgoing feeders for Capacitor Panel with each consisting of the following												
a)	1 No. 800 A, 415V, 65kA, 3 Pole, Electrically operated Draw Out type, ACB complete with safety shutters, 240V AC or 110V DC (control supply voltage to be selected in consultation with BMS Contractor) shunt trip coil & closing coil and having microprocessor based releases with (1) Protection against long time + short time + instantaneous + earth fault (all with adjustable time delays) (2) NO / NC contacts for interlocks and indications (3) 1 no CT in neutral to connect with the trip unit.												
b)	TNC switch												
c)	Selector switch for auto/manual operation.												
d)	LED indication lamp for ON, OFF & TRIP.												
e)	Digital Ammeter with Inbuilt Selector Switch												
f)	3 Nos. CTs 800/5A, 15VA, CL-1 for metering.												
g)	Power & control terminals for cable connections.												
7)	4Nos. 100 A, 415V, 50KA, 3P MCCB having Electronic Trip Unit with adjustable protection against Overload, adjustable protection against Short Circuit.(For DB-330, LDB-AB, SPDB-AB & 1No. Spare).												
a)	1 No. Digital Ammeter, CT operated.												
b)	1 No. CT 100/5A, 15VA, class - I for metering.												
c)	LED indication lamp for ON, OFF, Trip.												
8)	4 Nos. 400 A, 415V, 50KA, 3P MCCB having Electronic Trip Unit for Fire Pump Panel, Air Cooled Chiller Panel, Pump Room Panel & 1 No. spare with adjustable protection against Overload, adjustable protection against Short Circuit.												
a)	1 No. Digital Ammeter, CT operated.												
b)	1 No. CT 400/5A, 15VA, class - I for metering.												
c)	LED indication lamp for ON, OFF, Trip.												

SCHEDULE OF QUANTITIES FOR PART-A Electrical WORKS - CE-10 and 11 LOT-4(ADDENDUM)

S.No.	Description	Units	IGD Airport	mid shaft	Palam	Dashrath Puri	Dabri Mor	Janakpur West	Total Quantity	Rate in INR	Rate in Foreign Currency	Amount in INR	Amount in Foreign Currency
1.03.2	Delivery of equipment to site	set											
1.03.3	Installation and Equipment testing	set								44130			
1.03.4	System testing, Commissioning and hand over	set								2520			
1.03.5	Interlocking Arrangement	set								1260			
1.03.6	Scada\ BMS Connectivity at the separate Marshelling box in the panel with provision of all necessary I/O ponts & tranducers mounting prewired upto Marshelling box terminal boards from the factory itself and audited in factory from the BMS provider.	set								60			
1.04	Sub-Distribution Boards, DB-330 as per M & W Specifications , as per tender drawings Complete and as per the following details.	Sets	1	1	1	1	1	1	6	761210		4567260	
1.04.1													
A.	INCOMER												
a)	1 No. 100 A, 415V, 50KA, 3P MCCB having Electronic Trip Unit with												
b)	LED indication lamp for the following- ON, OFF, TRIP (each for both the supplies)												
c)	LED phase indication lamp for R-Y-B.												
d)	1 No Digital Ammeter with Inbuilt Selector Switch.												
e)	3 Nos. CTs, 100/5A, 15VA, CL-1 for metering.												
B.	BUS BAR												
	Electrolytic high conductivity T P & N (full size) copper conductor bus-bars												
C.	OUTGOING												
1)	4 Sets of Star - Delta Stater unit (For VSF & VEF) of up to 7.5 kW and each shall be equipped with the followings but not limited to:												
a)	1 No. 3 Pole suitable rating MPCB (with Inbuilt Single Phasing Preventor)												
b)	1 No. Digital Ammeter, CT operated.												
c)	1 No. CT 5A, 15VA, class - I for metering.												
d)	Local / Remote Selector Switch key operated.												
e)	Door mounted push buttons for Start and Stop functions.												
f)	Emergency stop push button lockable type.												
g)	LED indication lamp for ON, OFF, Trip.												
h)	3 No. with suitable rating , AC-3 Duty, 3 - pole Contactor with Auxiliary												
i)	One no Star-Delta timer.												
j)	1 No. Auxiliary Contactor with 4 Nos. Auxiliary Contacts for Remote												
k)	Power & control terminals for cable connections.												
2)	6 Sets of Star - Delta Stater unit (For VSF & VEF & 1 No. spare) of up to 3.7 kW and each shall be equipped with the followings but not limited to:												
a)	1 No. 3 Pole suitable rating MPCB (with Inbuilt Single Phasing Preventor) with Over - load release and having contacts for indications - On, Off, Trip.												
b)	1 No. Digital Ammeter, CT operated.												
c)	1 No. CT 5A, 15VA, class - I for metering.												
d)	Local / Remote Selector Switch key operated.												
e)	Door mounted push buttons for Start and Stop functions.												
f)	Emergency stop push button lockable type.												
g)	LED indication lamp for ON, OFF, Trip.												
h)	3 No. with suitable rating , AC-3 Duty, 3 - pole Contactor with Auxiliary Contacts and Operating Coil of 230 V / 110 V AC.(for Main, Star & Delta Contactor).												
i)	One no Star-Delta timer.												

SCHEDULE OF QUANTITIES FOR PART-A Electrical WORKS - CE-10&11 LOT-4(ADDENDUM)

S.No.	Description	Units	IGD Airport	mid shaft	Palam	Deshrath Pun	Dabri Mor	Janakpur West	Total Quantity	Rate in INR	Rate in Foreign Currency	Amount in INR	Amount in Foreign Currency
j)	1 No. Auxiliary Contactor with 4 Nos. Auxiliary Contacts for Remote Operation.												
k)	Power & control terminals for cable connections.												
	14 Nos 0.5 kW single phase feeder for Motor Operated												
1.04.2	Delivery of equipment to site	set	1	1	1	1	1	1	6				
1.04.3	Installation and Equipment testing	set	1	1	1	1	1	1	6	21440		128640	
1.04.4	System testing, Commissioning and hand over	set	1	1	1	1	1	1	6	2520		15120	
1.04.5	Interlocking Arrangement	set	1	1	1	1	1	1	6	2520		15120	
1.04.6	Scada\ BMS Connectivity at the separate Marshelling box in the panel with provision of all necessary I/O points & transducers mounting prewired upto Marshelling box terminal boards from the factory itself and audited in factory from the BMS provider.	set	1	1	1	1	1	1	6	1260		7560	
1.05	Main Distribution Board (MDB), DB-130 & DB-230 as per M & W Specifications, as per tender drawings Complete & as per following details:	Sets	3		2	2	2	3	12	4724580		56694960	
1.05.1													
A	INCOMER												
1	Automatic transfer switch with 2 Nos 1000A, 4 pole, 65 50 kA electric drawout (EDO) ACBs, electrically and mechanically interlocked , controller having selector switch for automatic & forced operation on normal and emergency sources, monitoring of normal source and automatic changeover, test facility, circuit breaker status indication (On, Off & Trip), 3 ph Under - voltage protection, Phase Sequence Protection inbuilt in the controller or separately provided, the controller should have communication capability with BMS/SCADA. The ACB complete with safety shutters, 240V AC or 110V DC (control supply voltage to be selected in consultation with BMS Contractor) shunt trip coil & closing coil and having microprocessor based releases with (1) Communication capability (2) Provision for measurement of three phase current (3) Protection against long time + short time + instantaneous + earth fault (all with adjustable time delays), Energy metering (4) Zone Selective Interlocking Unit for Total Discrimination (5) Commuto remotely read and set parameters for the protection functions, transmission												
	of ammeter measurements (6) Signalling of the cause of tripping and alarms, maximeter reset etc (7) NO / NC contacts for interlocks and indications (8) 1 no CT in neutral to connect with the trip unit. must have Electronic Trip Unit with Overload & Short Circuit Protection, closing & shunt trip coil, auxiliary contacts for ON, OFF, TRIP indication etc. as per drawing and specification.												
a)	LED indication lamp for the following- ON, OFF, TRIP (each for both the supplies)												
b)	LED phase indication lamp for R-Y-B.												
c)	Metering												
i)	Multifunction energy meter for measurement of voltage, line current, power factor, power & energy.												
ii)	3 Nos. CTs 1000/5A, 15VA, CL-1 for metering.												
iii)	1 Nos. CTs 1000/5A, 15VA, CL-1 for APFCR relay.												
B.	BUS BAR												
	Electrolytic high conductivity T P & N+E copper conductor bus-bars (as specified in the Specifications) rated 1250 Amps, suitable to withstand symmetrical fault level of 56 50KA for 1 second at 415 volts.												